STA 522 Project 1

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Contents

1. Data Set

Data Source

This project uses data from the College Scorecard, available at https://collegescorecard.ed.gov/. We use two main datasets:

1. **Most-Recent-Cohorts-Institution_05192025.csv**: Institution-level data
2. **Most-Recent-Cohorts-Field-of-Study.csv**: Program-level data for specific majors

Variables Used

The following table summarizes the variables used across all four problems:

Variable	Description	Used in Problem(s)
UNITID	Unique institution identifier	1, 2, 3, 4
INSTNM	Institution name	1, 2, 3, 4
UGDS	Undergraduate enrollment (target for estimation)	1
CONTROL	Control type: 1 = Public, 2 = Private nonprofit, 3 = Private for-profit	1, 2, 3, 4
ICLEVEL	Institutional level: 1 = 4+ years, 2 = 2-4 years, 3 = <2 years	1, 2, 3, 4
PREDDEG	Predominant degree: 0 = Non-degree, 1 = Certificate, 2 = Associate, 3 = Bachelor, 4 = Graduate only	1, 2, 3, 4
CURROPER	Operating status: 1 = currently operating	1, 2, 3, 4
MD_EARN_WNE_P10	Median alumni earnings 10 years after entry (in dollars)	3
COSTT4_A	Average annual cost of attendance (in dollars)	4
CIPCODE	Classification of Instructional Programs code (2705 = Statistics, 2706 = Mathematical Statistics)	2
CREDDESC	Award level (Undergraduate Certificate, Associate's Degree, Bachelor's Degree, etc.)	2

2. Survey Design

Objective

This survey design is used for all four estimation problems:

1. **Problem 1**: Estimate total undergraduate enrollment across U.S. institutions
2. **Problem 2**: Estimate proportion of institutions offering undergraduate Statistics major
3. **Problem 3**: Estimate domain means of median alumni earnings for Public vs. Private institutions
4. **Problem 4**: Estimate domain means of average annual cost for Public vs. Private institutions

Population and Sampling Frame

Target Population: Active U.S. institutions offering undergraduate education

Sampling Frame: The frame consists of institutions from the College Scorecard data satisfying:

1. CURROPER == 1 (currently operating)
2. UGDS > 0 or PREDDEG in {1,2,3} (offers undergraduate education)
3. Non-missing UGDS (has valid enrollment data)

Stratified Random Sampling Design

Stratification

We use a **stratified random sampling** design with **six strata** defined by crossing two factors:

- **Control type**: Public (CONTROL = 1) vs. Private (CONTROL in {2,3})
- **Institutional level (ICLEVEL)**:
  - ICLEVEL = 1: Four or more years (4-year institutions)
  - ICLEVEL = 2: At least 2 but less than 4 years (2-year institutions)
  - ICLEVEL = 3: Less than 2 years (certificate-granting institutions)

This creates six strata:

1. **Public-ICLEVEL1**: Public 4-year+ institutions
2. **Public-ICLEVEL2**: Public 2-year institutions
3. **Public-ICLEVEL3**: Public certificate-granting institutions (< 2 years)
4. **Private-ICLEVEL1**: Private 4-year+ institutions
5. **Private-ICLEVEL2**: Private 2-year institutions
6. **Private-ICLEVEL3**: Private certificate-granting institutions (< 2 years)

Rationale for Stratification

Control type (public vs. private) and institutional level (ICLEVEL: 4-year+, 2-year, <2 years) are key drivers of institutional characteristics. This 2×3 stratification captures major sources of variation in enrollment, program offerings, alumni earnings, and costs while maintaining adequate sample sizes.

Sample Size and Allocation

```
## Stratum sizes and sample allocation:

## # A tibble: 6 x 4
##   stratum            N_h   n_h sampling_rate
##   <chr>            <int> <dbl> <chr>
## 1 Private-ICLEVEL1  1653    59 3.57%
## 2 Private-ICLEVEL3  1390    50 3.60%
## 3 Public-ICLEVEL2    867    31 3.58%
## 4 Public-ICLEVEL1    806    29 3.60%
## 5 Private-ICLEVEL2   637    23 3.61%
## 6 Public-ICLEVEL3    224     8 3.57%
```

Total sample size: $n = 200$ institutions

Allocation method: **Proportional allocation** where $n_h \propto N_h$, ensuring each stratum is represented proportionally to its population size. This approach:

- Maximizes precision for population-level estimates when variance is roughly constant across strata
- Maintains self-weighting property within each stratum
- Ensures adequate sample sizes in all strata

Sampling Method

Within each stratum h:

- **Method**: Simple random sampling without replacement
- **Sample size**: n_h institutions selected from N_h total
- **Sampling weight**: Each sampled institution i in stratum h receives weight $w_i = \frac{N_h}{n_h}$

Sample Selection

```
##
## Total sample size: 200 institutions
```

This sampling design is used consistently across all four estimation problems, allowing for:

- Efficient estimation of multiple parameters from a single sample
- Consistent stratification that captures key sources of variation
- Valid design-based inference for totals, proportions, and domain means

3. Estimation

3.1 Problem 1: Total Undergraduate Enrollment

Question

What is an estimate of the total number of undergraduate students enrolled in colleges?

Parameter of Interest

Estimate the total undergraduate enrollment across U.S. institutions:

$$T = \sum_{i=1}^{N} y_i$$

where y_i is the undergraduate enrollment (UGDS) at institution i.

Estimation Method

Under stratified random sampling, the **stratified expansion estimator** is:

$$\hat{T} = \sum_{h=1}^{H} N_h \bar{y}_h$$

where:

- $H = 6$ is the number of strata
- N_h is the population size of stratum h
- $\bar{y}_h = \frac{1}{n_h} \sum_{i \in s_h} y_i$ is the sample mean in stratum h

Variance and Confidence Interval

The variance estimator with finite population correction (FPC) is:

$$\widehat{\text{Var}}(\hat{T}) = \sum_{h=1}^{H} N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}$$

where $s_h^2 = \frac{1}{n_h - 1} \sum_{i \in s_h} (y_i - \bar{y}_h)^2$ is the sample variance in stratum h.

A **95% confidence interval** is constructed as:

$$\hat{T} \pm 1.96\sqrt{\widehat{\text{Var}}(\hat{T})}$$

assuming approximate normality by the Central Limit Theorem.

Results

```r
# Calculate stratum-level estimates
stratum_estimates <- sample_data %>%
  group_by(stratum) %>%
  summarise(
    n_h = n(),
    y_bar_h = mean(UGDS),
    s_h_sq = var(UGDS),
    .groups = 'drop'
  ) %>%
  left_join(stratum_N, by = 'stratum')

cat('Stratum-level estimates:\n')
```

```
## Stratum-level estimates:
```

```r
print(stratum_estimates %>%
        mutate(across(where(is.numeric), ~round(., 2))))
```

```
## # A tibble: 6 x 5
##   stratum            n_h y_bar_h    s_h_sq   N_h
##   <chr>            <dbl>   <dbl>     <dbl> <dbl>
## 1 Private-ICLEVEL1    59   1981. 14393687.  1653
## 2 Private-ICLEVEL2    23    346.   255212.   637
## 3 Private-ICLEVEL3    50    159.    21152.  1390
## 4 Public-ICLEVEL1     29   7088. 64301590.   806
## 5 Public-ICLEVEL2     31   4372  26267174.   867
## 6 Public-ICLEVEL3      8    495.   744184.   224
```

```r
# Total estimate
T_hat <- sum(stratum_estimates$N_h * stratum_estimates$y_bar_h)

# Variance estimate
var_T_hat <- sum(
  stratum_estimates$N_h^2 *
  (1 - stratum_estimates$n_h / stratum_estimates$N_h) *
  stratum_estimates$s_h_sq / stratum_estimates$n_h
)

se_T_hat <- sqrt(var_T_hat)
ci_lower <- T_hat - 1.96 * se_T_hat
ci_upper <- T_hat + 1.96 * se_T_hat

cat('\n=== FINAL RESULTS ===\n')
```

```
##
## === FINAL RESULTS ===
```

```r
cat('Estimated Total Undergraduate Enrollment:\n')
```

```
## Estimated Total Undergraduate Enrollment:
```

```r
cat('  Point Estimate: ', format(round(T_hat), big.mark=','), ' students\n', sep='')
```

```
##   Point Estimate: 13,330,475 students
```

```r
cat('  Standard Error: ', format(round(se_T_hat), big.mark=','), '\n', sep='')
```

```
##   Standard Error: 1,629,478
```

```r
cat('  95% Confidence Interval: [',
    format(round(ci_lower), big.mark=','), ', ',
    format(round(ci_upper), big.mark=','), ']\n', sep='')
```

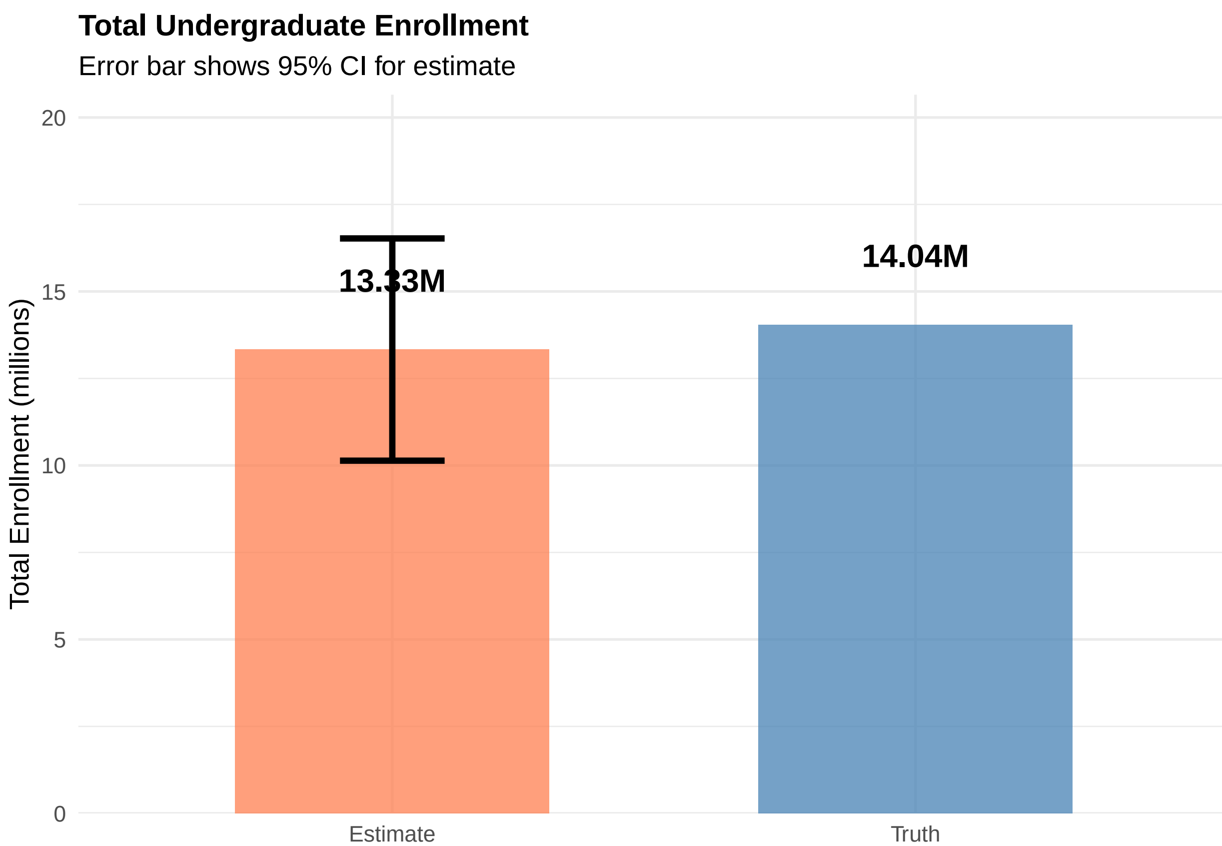
```
##   95% Confidence Interval: [10,136,698, 16,524,252]
```

Answer

Based on the stratified random sample of **200 institutions**, the estimated total number of undergraduate students enrolled in U.S. colleges is **13,330,475** students, with a 95% confidence interval of [10,136,698, 16,524,252].

Visualization

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.
```


Total Undergraduate Enrollment
Error bar shows 95% CI for estimate
Total Enrollment (millions)
20
15
10
5
0
13.33M
14.04M
Estimate
Truth

3.2 Problem 2: Proportion with Statistics Major

Question

What fraction of colleges have a major in statistical science? Use your sample to estimate this fraction; do not get the answer by filtering the schools.

Parameter of Interest

Estimate the proportion of U.S. colleges offering an undergraduate Statistics major:

$$P = \Pr(\text{institution offers undergraduate Statistics})$$

Response Variable

Binary indicator z_i for each institution i:

$$z_i = \begin{cases} 1 & \text{if institution } i \text{ offers undergraduate Statistics major} \\ 0 & \text{otherwise} \end{cases}$$

Operational Definition: "Statistics Major"

Using data from **Most-Recent-Cohorts-Field-of-Study.csv**, we define $z_i = 1$ if the institution has **at least one record** satisfying **both** conditions:

1. **Field of Study**: CIPCODE in {2705, 2706}, where:
   - 2705: Statistics (including General Statistics and Applied Statistics)
   - 2706: Mathematical Statistics and related programs
2. **Award Level**: The program is at the undergraduate level, indicated by `CREDDESC` containing:
   - "Undergraduate Certificate or Diploma"
   - "Associate's Degree"
   - "Bachelor's Degree"

Otherwise, set $z_i = 0$.

Estimation Method

We reuse the stratified random sample from Problem 1. The **stratified proportion estimator** is:

$$\hat{P} = \sum_{h=1}^{H} \frac{N_h}{N} \hat{p}_h$$

where:

- $\hat{p}_h = \frac{1}{n_h} \sum_{i \in s_h} z_i$ is the sample proportion in stratum h
- $N = \sum_{h=1}^{H} N_h$ is the total population size

Variance and Confidence Interval

The variance estimator is:

$$\widehat{\text{Var}}(\hat{P}) = \sum_{h=1}^{H} \left(\frac{N_h}{N}\right)^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}$$

where $s_h^2 = \dfrac{1}{n_h - 1} \sum_{i \in s_h} (z_i - \hat{p}_h)^2$ is the sample variance in stratum h.

A **95% confidence interval** is:

$$\hat{P} \pm 1.96\sqrt{\widehat{\text{Var}}(\hat{P})}$$

Results

```r
# Read field-of-study data
fos_data <- read.csv('Most-Recent-Cohorts-Field-of-Study.csv')

# Define undergraduate award levels
undergrad_levels <- c("Undergraduate Certificate or Diploma",
                      "Associate's Degree",
                      "Bachelor's Degree")

# Identify institutions with undergraduate Statistics majors
stats_institutions <- fos_data %>%
  filter(
    CIPCODE %in% c(2705, 2706),           # Statistics-related CIP codes
    CREDDESC %in% undergrad_levels         # Undergraduate level
  ) %>%
  distinct(UNITID) %>%
  mutate(has_stats = 1)

cat('Institutions with undergraduate Statistics programs in field-of-study data:',
    nrow(stats_institutions), '\n\n')
```

```
## Institutions with undergraduate Statistics programs in field-of-study data: 227
```

```r
# Join with sampled data to create indicator variable z_i
sample_with_indicator <- sample_data %>%
  left_join(stats_institutions, by = 'UNITID') %>%
  mutate(z_i = ifelse(is.na(has_stats), 0, 1))

# Check how many sampled institutions have Statistics majors
cat('Sampled institutions with undergraduate Statistics majors:\n')
```

```
## Sampled institutions with undergraduate Statistics majors:
```

```
cat('  Total:', sum(sample_with_indicator$z_i), 'out of', nrow(sample_with_indicator), '\n\n')
```

```
##   Total: 8 out of 200
```

```
# Calculate stratum-level proportions
stratum_proportions <- sample_with_indicator %>%
  group_by(stratum) %>%
  summarise(
    n_h = n(),
    z_sum = sum(z_i),
    p_hat_h = mean(z_i),
    s_h_sq = var(z_i),
    .groups = 'drop'
  ) %>%
  left_join(stratum_N, by = 'stratum')

cat('Stratum-level proportions:\n')
```

```
## Stratum-level proportions:
```

```
print(stratum_proportions %>%
        mutate(across(where(is.numeric), ~round(., 4))))
```

```
## # A tibble: 6 x 6
##   stratum             n_h z_sum p_hat_h s_h_sq   N_h
##   <chr>             <dbl> <dbl>   <dbl>  <dbl> <dbl>
## 1 Private-ICLEVEL1    59      1  0.0169 0.0169  1653
## 2 Private-ICLEVEL2    23      0  0       0       637
## 3 Private-ICLEVEL3    50      0  0       0      1390
## 4 Public-ICLEVEL1     29      7  0.241  0.190    806
## 5 Public-ICLEVEL2     31      0  0       0       867
## 6 Public-ICLEVEL3      8      0  0       0       224
```

```
# Calculate overall proportion estimate
N_total_pop <- sum(stratum_N$N_h)
P_hat <- sum((stratum_proportions$N_h / N_total_pop) * stratum_proportions$p_hat_h)

# Calculate variance
var_P_hat <- sum(
  (stratum_proportions$N_h / N_total_pop)^2 *
  (1 - stratum_proportions$n_h / stratum_proportions$N_h) *
  stratum_proportions$s_h_sq / stratum_proportions$n_h
)

se_P_hat <- sqrt(var_P_hat)
ci_lower_p <- P_hat - 1.96 * se_P_hat
ci_upper_p <- P_hat + 1.96 * se_P_hat

cat('\n=== FINAL RESULTS ===\n')
```

```
##
## === FINAL RESULTS ===
```

```r
cat('Estimated Proportion of Institutions with Undergraduate Statistics Major:\n')
```

```
## Estimated Proportion of Institutions with Undergraduate Statistics Major:
```

```r
cat('  Point Estimate: ', sprintf('%.4f', P_hat), ' (', sprintf('%.2f%%', P_hat * 100), ')\n', sep='')
```

```
##   Point Estimate: 0.0399 (3.99%)
```

```r
cat('  Standard Error: ', sprintf('%.4f', se_P_hat), '\n', sep='')
```

```
##   Standard Error: 0.0125
```

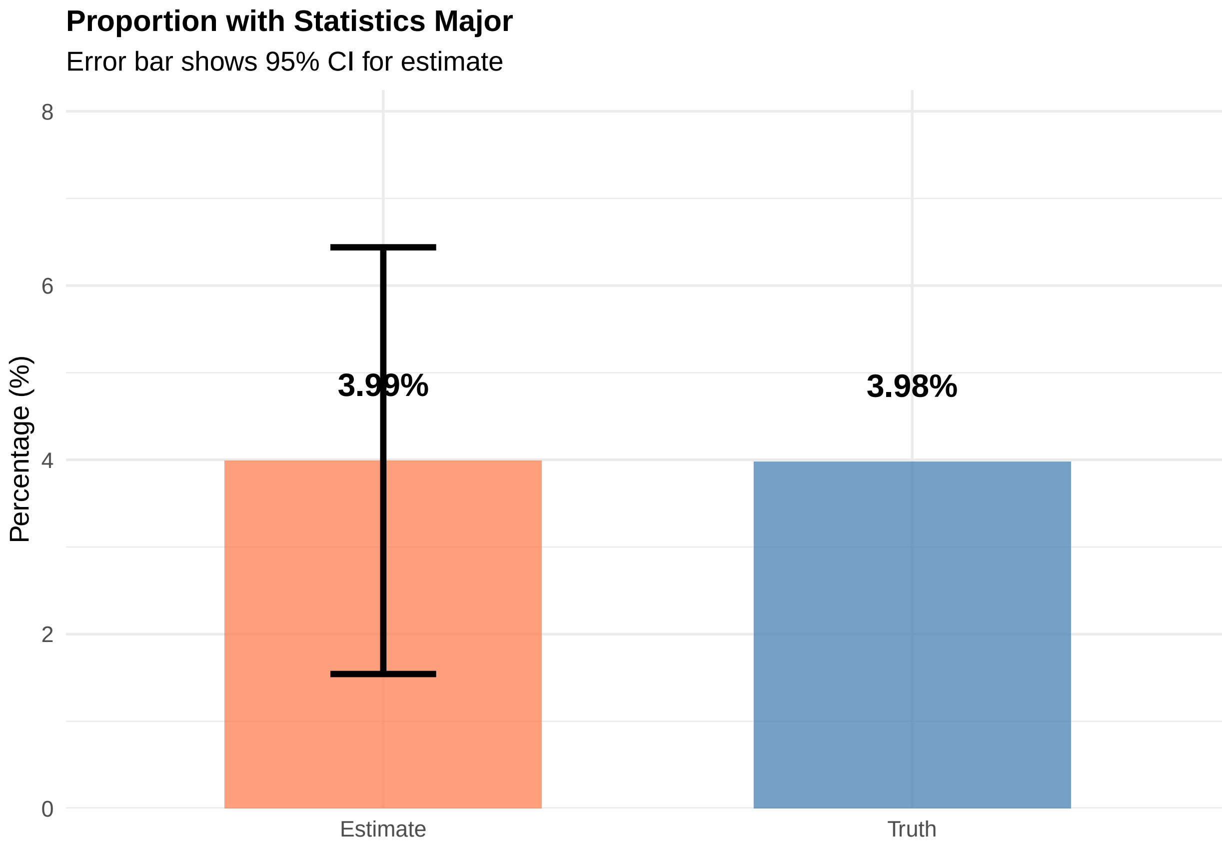
```r
cat('  95% Confidence Interval: [',
    sprintf('%.4f', ci_lower_p), ', ',
    sprintf('%.4f', ci_upper_p), '] or [',
    sprintf('%.2f%%', ci_lower_p * 100), ', ',
    sprintf('%.2f%%', ci_upper_p * 100), ']\n', sep='')
```

```
##   95% Confidence Interval: [0.0154, 0.0644] or [1.54%, 6.44%]
```

Answer

Based on the stratified random sample of **200 institutions**, the estimated proportion of U.S. colleges offering an undergraduate Statistics major is **3.99%**, with a 95% confidence interval of [1.54%, 6.44%].

Visualization



3.3 Problem 3: Alumni Earnings by Control Type

Question

What is the average of the median earnings among alumni from public schools? From private schools (combine for profit and not for profit private schools)?

Parameter of Interest

Estimate the **domain means** of median alumni earnings for:

- **Public institutions**: $\mu_{\text{Public}} = E[y_i | D_i = \text{Public}]$
- **Private institutions**: $\mu_{\text{Private}} = E[y_i | D_i = \text{Private}]$

where $y_i = \text{MD\_EARN\_WNE\_P10}$ is the median earnings of alumni 10 years after entry (in dollars).

Domain Classification

- $D_i = \text{Public if CONTROL} = 1$
- $D_i = \text{Private if CONTROL in } \{2, 3\}$ (combining nonprofit and for-profit private institutions)

Missing Data

If y_i is missing or suppressed (e.g., "PrivacySuppressed"), exclude institution i from both numerator and denominator for that domain. This maintains design-unbiasedness under the missing-at-random (MAR) assumption within strata.

Estimation Method

We reuse the stratified random sample from Problem 1. For each domain $d \in \{\text{Public}, \text{Private}\}$, the **weighted ratio estimator** (domain mean under stratified sampling) is:

$$\hat{\mu}_d = \frac{\sum_{i \in s} w_i y_i \mathbf{1}(D_i = d)}{\sum_{i \in s} w_i \mathbf{1}(D_i = d)}$$

where:

- $w_i = \dfrac{N_h}{n_h}$ is the sampling weight for institution i in stratum h
- $\mathbf{1}(D_i = d)$ is the indicator function equal to 1 if institution i belongs to domain d

Equivalently, expressed by strata:

$$\hat{\mu}_d = \frac{\sum_{h=1}^{H} \sum_{i \in s_h} w_i y_i \mathbf{1}(D_i = d)}{\sum_{h=1}^{H} \sum_{i \in s_h} w_i \mathbf{1}(D_i = d)}$$

Variance Estimation

We use the **survey package** in R for proper domain estimation under stratified sampling. The package implements the correct variance estimator for domain means.

For each stratum h and domain d, let:

- n_h = total number of sampled institutions in stratum h (from original design)
- n_{hd} = number of sampled institutions in stratum h belonging to domain d with non-missing y_i
- s_{hd}^2 = sample variance of y_i among the n_{hd} institutions
- $W_{hd} = \sum_{i \in s_h} w_i \mathbf{1}(D_i = d)$ = sum of weights for domain d in stratum h

The variance estimator is:

$$\widehat{\text{Var}}(\hat{\mu}_d) = \sum_{h=1}^{H} \left(\frac{W_{hd}}{\sum_{h'=1}^{H} W_{h'd}} \right)^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_{hd}^2}{n_{hd}}$$

Key point: In the finite population correction (FPC) term, we use n_h/N_h (the overall stratum sampling rate), not n_{hd}/N_h (the domain-specific sample size), because the sample was selected from the entire stratum, not from the domain subset. This is automatically handled correctly by the `survey` package using `svydesign()` with `fpc = ~N_h` and `svyby()` for domain estimation.

A **95% confidence interval** is:

$$\hat{\mu}_d \pm 1.96\sqrt{\widehat{\text{Var}}(\hat{\mu}_d)}$$

Results

```
# Add earnings data to sample
sample_with_earnings <- sample_data %>%
  mutate(
    domain = ifelse(CONTROL == 1, 'Public', 'Private'),
    earnings = suppressWarnings(as.numeric(MD_EARN_WNE_P10))
  )

# Check for missing/suppressed earnings
cat('=== DATA QUALITY CHECK ===\n')
```

```
## === DATA QUALITY CHECK ===
```

```
cat('Total sampled institutions:', nrow(sample_with_earnings), '\n')
```

```
## Total sampled institutions: 200
```

```
cat('Missing or suppressed earnings:', sum(is.na(sample_with_earnings$earnings)), '\n')
```

```
## Missing or suppressed earnings: 33
```

```r
cat('Valid earnings data:', sum(!is.na(sample_with_earnings$earnings)), '\n\n')
```

```
## Valid earnings data: 167
```

```r
# Summary by domain
cat('Breakdown by domain:\n')
```

```
## Breakdown by domain:
```

```r
domain_summary <- sample_with_earnings %>%
  group_by(domain) %>%
  summarise(
    n_total = n(),
    n_missing = sum(is.na(earnings)),
    n_valid = sum(!is.na(earnings)),
    .groups = 'drop'
  )
print(domain_summary)
```

```
## # A tibble: 2 x 4
##   domain  n_total n_missing n_valid
##   <chr>     <int>     <int>   <int>
## 1 Private     132        31     101
## 2 Public       68         2      66
```

```r
# Check for empty stratum-domain cells
cat('\n\nStratum-domain cell counts:\n')
```

```
##
##
## Stratum-domain cell counts:
```

```r
cell_counts <- sample_with_earnings %>%
  filter(!is.na(earnings)) %>%
  group_by(stratum, domain) %>%
  summarise(n_hd = n(), .groups = 'drop') %>%
  arrange(stratum, domain)
print(cell_counts)
```

```
## # A tibble: 6 x 3
##   stratum          domain   n_hd
##   <chr>            <chr>   <int>
## 1 Private-ICLEVEL1 Private    47
## 2 Private-ICLEVEL2 Private    19
## 3 Private-ICLEVEL3 Private    35
## 4 Public-ICLEVEL1  Public     29
## 5 Public-ICLEVEL2  Public     31
## 6 Public-ICLEVEL3  Public      6
```

```r
# === USING SURVEY PACKAGE FOR DOMAIN ESTIMATION ===
cat('\n=== DOMAIN MEAN ESTIMATION (using survey package) ===\n')
```

```
##
## === DOMAIN MEAN ESTIMATION (using survey package) ===
```

```r
# Add stratum info to sample data
sample_with_earnings <- sample_with_earnings %>%
  left_join(stratum_N, by = 'stratum')

# Create survey design object
# fpc = N_h for each stratum (finite population correction)
design_earnings <- svydesign(
  ids = ~1,                    # No clustering, simple random sampling within strata
  strata = ~stratum,           # Stratification variable
  fpc = ~N_h,                  # Finite population size for each stratum
  data = sample_with_earnings
)

# Perform domain estimation using svyby()
# This automatically handles the correct variance estimation for domains
domain_means <- svyby(
  formula = ~earnings,          # Variable to estimate
  by = ~domain,                 # Domain variable
  design = design_earnings,     # Survey design object
  FUN = svymean,                # Function to apply (mean)
  na.rm = TRUE,                 # Remove missing values
  keep.names = FALSE
)

cat('\nDomain estimates from svyby:\n')
```

```
##
## Domain estimates from svyby:
```

```r
print(domain_means)
```

```
##    domain earnings       se
## 1 Private  43797.9 1858.324
## 2  Public  45889.6 1306.064
```

```r
# Extract results
public_results <- list(
  mu_hat = domain_means$earnings[domain_means$domain == 'Public'],
  se = domain_means$se[domain_means$domain == 'Public'],
  n_total = sum(sample_with_earnings$domain == 'Public' & !is.na(sample_with_earnings$earnings))
)
public_results$var_mu_hat <- public_results$se^2
public_results$ci_lower <- public_results$mu_hat - 1.96 * public_results$se
public_results$ci_upper <- public_results$mu_hat + 1.96 * public_results$se
```

```r
private_results <- list(
  mu_hat = domain_means$earnings[domain_means$domain == 'Private'],
  se = domain_means$se[domain_means$domain == 'Private'],
  n_total = sum(sample_with_earnings$domain == 'Private' & !is.na(sample_with_earnings$earnings))
)
private_results$var_mu_hat <- private_results$se^2
private_results$ci_lower <- private_results$mu_hat - 1.96 * private_results$se
private_results$ci_upper <- private_results$mu_hat + 1.96 * private_results$se

# Display final results
cat('\n=== FINAL RESULTS ===\n\n')
```

```
##
## === FINAL RESULTS ===
```

```r
cat('PUBLIC INSTITUTIONS:\n')
```

```
## PUBLIC INSTITUTIONS:
```

```r
cat('  Sample size (with valid earnings):', public_results$n_total, '\n')
```

```
##   Sample size (with valid earnings): 66
```

```r
cat('  Estimated mean earnings: $', format(round(public_results$mu_hat), big.mark=','), '\n', sep='')
```

```
##   Estimated mean earnings: $45,890
```

```r
cat('  Standard error: $', format(round(public_results$se), big.mark=','), '\n', sep='')
```

```
##   Standard error: $1,306
```

```r
cat('  95% Confidence Interval: [$',
    format(round(public_results$ci_lower), big.mark=','), ', $',
    format(round(public_results$ci_upper), big.mark=','), ']\n\n', sep='')
```

```
##   95% Confidence Interval: [$43,330, $48,449]
```

```r
cat('PRIVATE INSTITUTIONS:\n')
```

```
## PRIVATE INSTITUTIONS:
```

```r
cat('  Sample size (with valid earnings):', private_results$n_total, '\n')
```

```
##   Sample size (with valid earnings): 101
```

```r
cat('  Estimated mean earnings: $', format(round(private_results$mu_hat), big.mark=','), '\n', sep='')
```

```
##   Estimated mean earnings: $43,798
```

```r
cat('  Standard error: $', format(round(private_results$se), big.mark=','), '\n', sep='')
```

```
##   Standard error: $1,858
```

```r
cat('  95% Confidence Interval: [$',
    format(round(private_results$ci_lower), big.mark=','), ', $',
    format(round(private_results$ci_upper), big.mark=','), ']\n\n', sep='')
```

```
##   95% Confidence Interval: [$40,156, $47,440]
```

```r
cat('DIFFERENCE (Private - Public):\n')
```

```
## DIFFERENCE (Private - Public):
```

```r
diff_estimate <- private_results$mu_hat - public_results$mu_hat
diff_se <- sqrt(private_results$var_mu_hat + public_results$var_mu_hat)
diff_ci_lower <- diff_estimate - 1.96 * diff_se
diff_ci_upper <- diff_estimate + 1.96 * diff_se

cat('  Point estimate: $', format(round(diff_estimate), big.mark=','), '\n', sep='')
```

```
##   Point estimate: $-2,092
```

```r
cat('  Standard error: $', format(round(diff_se), big.mark=','), '\n', sep='')
```

```
##   Standard error: $2,271
```

```r
cat('  95% CI: [$', format(round(diff_ci_lower), big.mark=','), ', $',
    format(round(diff_ci_upper), big.mark=','), ']\n', sep='')
```

```
##   95% CI: [$-6,544, $2,360]
```

```r
cat('  Statistically significant at  =0.05?',
    ifelse(diff_ci_lower > 0 | diff_ci_upper < 0, 'YES', 'NO'), '\n\n')
```

```
##   Statistically significant at  =0.05? NO
```

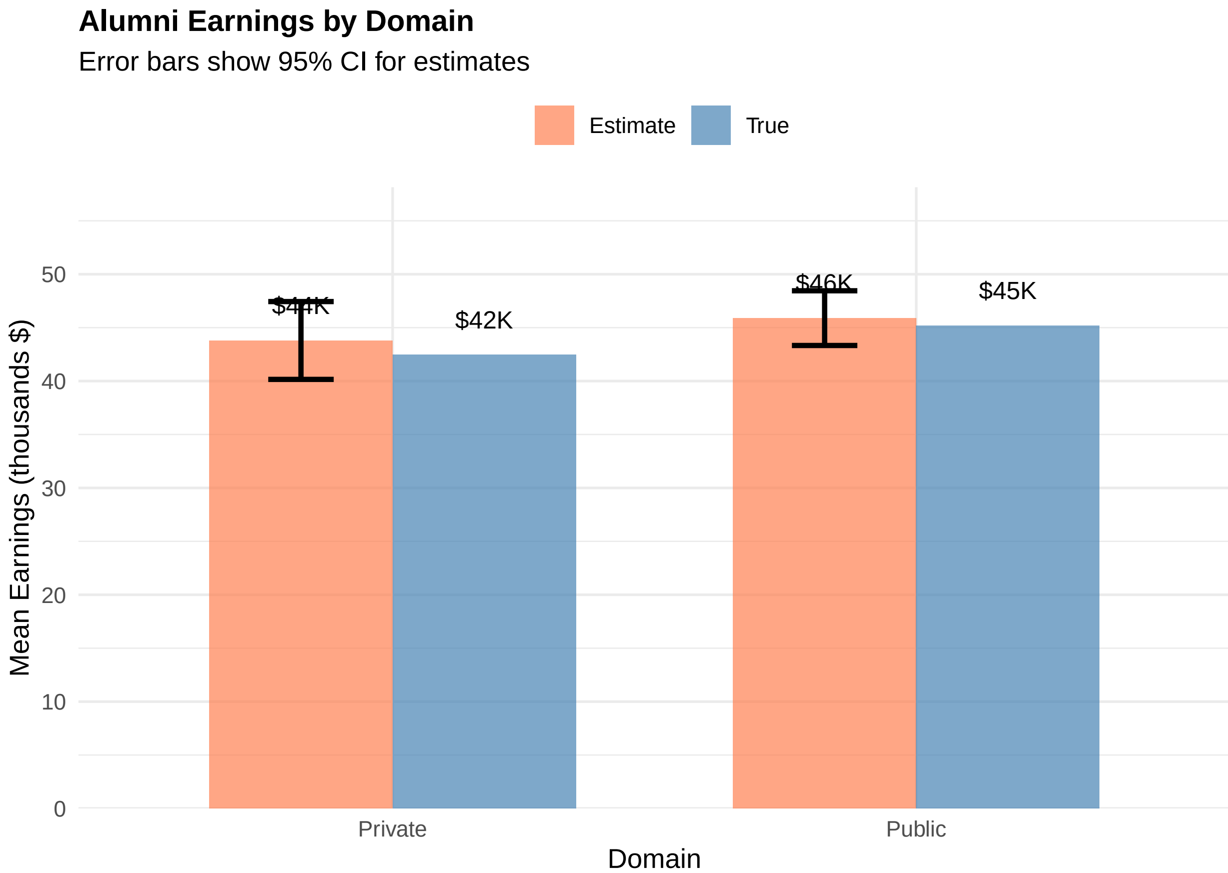
Answer

Based on the stratified random sample of **200 institutions**:

- The estimated average median earnings for **public institutions** is **$45,890**, with a 95% confidence interval of [$43,330,$48,449].
- The estimated average median earnings for **private institutions** is **$43,798**, with a 95% confidence interval of [$40,156,$47,440].

These estimates represent the design-weighted average of institutional-level median earnings, accounting for the stratified sampling design and unequal sampling rates across strata.

Visualization



3.4 Problem 4: Average Annual Cost by Control Type

Question

What is the average annual cost for public schools? For private schools (combine for profit and not for profit private schools)?

Parameter of Interest

Estimate the **domain means** of average annual cost for:

- **Public institutions**: $\mu_{\text{Public}} = E[y_i|D_i = \text{Public}]$
- **Private institutions**: $\mu_{\text{Private}} = E[y_i|D_i = \text{Private}]$

where $y_i = \text{COSTT4\_A}$ is the average annual cost of attendance (in dollars).

Domain Classification

Same as Problem 3.

Missing Data

Same as Problem 3.

Estimation Method

Same as Problem 3.

Variance Estimation

Same as Problem 3.

Results

```
# Add cost data to sample
sample_with_cost <- sample_data %>%
  mutate(
    domain = ifelse(CONTROL == 1, 'Public', 'Private'),
    cost = suppressWarnings(as.numeric(COSTT4_A))
  )

# Check for missing/suppressed cost data
cat('=== DATA QUALITY CHECK ===\n')
```

```
## === DATA QUALITY CHECK ===
```

```r
cat('Total sampled institutions:', nrow(sample_with_cost), '\n')
```

```
## Total sampled institutions: 200
```

```r
cat('Missing or suppressed cost data:', sum(is.na(sample_with_cost$cost)), '\n')
```

```
## Missing or suppressed cost data: 88
```

```r
cat('Valid cost data:', sum(!is.na(sample_with_cost$cost)), '\n\n')
```

```
## Valid cost data: 112
```

```r
# Summary by domain
cat('Breakdown by domain:\n')
```

```
## Breakdown by domain:
```

```r
domain_summary_cost <- sample_with_cost %>%
  group_by(domain) %>%
  summarise(
    n_total = n(),
    n_missing = sum(is.na(cost)),
    n_valid = sum(!is.na(cost)),
    pct_missing = sprintf('%.1f%%', sum(is.na(cost)) / n() * 100),
    .groups = 'drop'
  )
print(domain_summary_cost)
```

```
## # A tibble: 2 x 5
##   domain  n_total n_missing n_valid pct_missing
##   <chr>     <int>     <int>   <int> <chr>
## 1 Private     132        76      56 57.6%
## 2 Public       68        12      56 17.6%
```

```r
# Check for empty stratum-domain cells
cat('\n\nStratum-domain cell counts:\n')
```

```
##
##
## Stratum-domain cell counts:
```

```r
cell_counts_cost <- sample_with_cost %>%
  filter(!is.na(cost)) %>%
  group_by(stratum, domain) %>%
  summarise(n_hd = n(), .groups = 'drop') %>%
  arrange(stratum, domain)
print(cell_counts_cost)
```

```
## # A tibble: 6 x 3
##   stratum          domain   n_hd
##   <chr>            <chr>   <int>
## 1 Private-ICLEVEL1 Private    49
## 2 Private-ICLEVEL2 Private     6
## 3 Private-ICLEVEL3 Private     1
## 4 Public-ICLEVEL1  Public     27
## 5 Public-ICLEVEL2  Public     28
## 6 Public-ICLEVEL3  Public      1
```

```r
# === USING SURVEY PACKAGE FOR DOMAIN ESTIMATION ===
cat('\n=== DOMAIN MEAN ESTIMATION (using survey package) ===\n')
```

```
##
## === DOMAIN MEAN ESTIMATION (using survey package) ===
```

```r
# Add stratum info to sample data
sample_with_cost <- sample_with_cost %>%
  left_join(stratum_N, by = 'stratum')

# Create survey design object
# fpc = N_h for each stratum (finite population correction)
design_cost <- svydesign(
  ids = ~1,                      # No clustering, simple random sampling within strata
  strata = ~stratum,             # Stratification variable
  fpc = ~N_h,                    # Finite population size for each stratum
  data = sample_with_cost
)

# Perform domain estimation using svyby()
# This automatically handles the correct variance estimation for domains
domain_costs <- svyby(
  formula = ~cost,                # Variable to estimate
  by = ~domain,                   # Domain variable
  design = design_cost,           # Survey design object
  FUN = svymean,                  # Function to apply (mean)
  na.rm = TRUE,                   # Remove missing values
  keep.names = FALSE
)

cat('\nDomain estimates from svyby:\n')
```

```
##
## Domain estimates from svyby:
```

```r
print(domain_costs)
```

```
##    domain     cost        se
## 1 Private 40713.44 2393.9858
## 2  Public 18712.65  694.8756
```

```
# Extract results
public_cost_results <- list(
  mu_hat = domain_costs$cost[domain_costs$domain == 'Public'],
  se = domain_costs$se[domain_costs$domain == 'Public'],
  n_total = sum(sample_with_cost$domain == 'Public' & !is.na(sample_with_cost$cost))
)
public_cost_results$var_mu_hat <- public_cost_results$se^2
public_cost_results$ci_lower <- public_cost_results$mu_hat - 1.96 * public_cost_results$se
public_cost_results$ci_upper <- public_cost_results$mu_hat + 1.96 * public_cost_results$se

private_cost_results <- list(
  mu_hat = domain_costs$cost[domain_costs$domain == 'Private'],
  se = domain_costs$se[domain_costs$domain == 'Private'],
  n_total = sum(sample_with_cost$domain == 'Private' & !is.na(sample_with_cost$cost))
)
private_cost_results$var_mu_hat <- private_cost_results$se^2
private_cost_results$ci_lower <- private_cost_results$mu_hat - 1.96 * private_cost_results$se
private_cost_results$ci_upper <- private_cost_results$mu_hat + 1.96 * private_cost_results$se

# Display final results
cat('\n=== FINAL RESULTS ===\n\n')
```

```
##
## === FINAL RESULTS ===
```

```
cat('PUBLIC INSTITUTIONS:\n')
```

```
## PUBLIC INSTITUTIONS:
```

```
cat('  Sample size (with valid cost data):', public_cost_results$n_total, '\n')
```

```
##   Sample size (with valid cost data): 56
```

```
cat('  Estimated mean annual cost: $', format(round(public_cost_results$mu_hat), big.mark=','), '\n', se
```

```
##   Estimated mean annual cost: $18,713
```

```
cat('  Standard error: $', format(round(public_cost_results$se), big.mark=','), '\n', sep='')
```

```
##   Standard error: $695
```

```
cat('  95% Confidence Interval: [$',
    format(round(public_cost_results$ci_lower), big.mark=','), ', $',
    format(round(public_cost_results$ci_upper), big.mark=','), ']\n\n', sep='')
```

```
##   95% Confidence Interval: [$17,351, $20,075]
```

```
cat('PRIVATE INSTITUTIONS:\n')
```

```
## PRIVATE INSTITUTIONS:
```

```
cat('  Sample size (with valid cost data):', private_cost_results$n_total, '\n')
```

```
##   Sample size (with valid cost data): 56
```

```
cat('  Estimated mean annual cost: $', format(round(private_cost_results$mu_hat), big.mark=','), '\n', 
```

```
##   Estimated mean annual cost: $40,713
```

```
cat('  Standard error: $', format(round(private_cost_results$se), big.mark=','), '\n', sep='')
```

```
##   Standard error: $2,394
```

```
cat('  95% Confidence Interval: [$',
    format(round(private_cost_results$ci_lower), big.mark=','), ', $',
    format(round(private_cost_results$ci_upper), big.mark=','), ']\n\n', sep='')
```

```
##   95% Confidence Interval: [$36,021, $45,406]
```

```
cat('DIFFERENCE (Private - Public):\n')
```

```
## DIFFERENCE (Private - Public):
```

```
diff_cost_estimate <- private_cost_results$mu_hat - public_cost_results$mu_hat
diff_cost_se <- sqrt(private_cost_results$var_mu_hat + public_cost_results$var_mu_hat)
diff_cost_ci_lower <- diff_cost_estimate - 1.96 * diff_cost_se
diff_cost_ci_upper <- diff_cost_estimate + 1.96 * diff_cost_se

cat('  Point estimate: $', format(round(diff_cost_estimate), big.mark=','), '\n', sep='')
```

```
##   Point estimate: $22,001
```

```
cat('  Standard error: $', format(round(diff_cost_se), big.mark=','), '\n', sep='')
```

```
##   Standard error: $2,493
```

```
cat('  95% CI: [$', format(round(diff_cost_ci_lower), big.mark=','), ', $',
    format(round(diff_cost_ci_upper), big.mark=','), ']\n', sep='')
```

```
##   95% CI: [$17,115, $26,887]
```

```r
cat('  Statistically significant at  =0.05?',
    ifelse(diff_cost_ci_lower > 0 | diff_cost_ci_upper < 0, 'YES', 'NO'), '\n\n')
```

```
##   Statistically significant at  =0.05? YES
```

Answer

Based on the stratified random sample of **200 institutions**:

- The estimated average annual cost for **public institutions** is **$18,713**, with a 95% confidence interval of [$17,351,$20,075].
- The estimated average annual cost for **private institutions** is **$40,713**, with a 95% confidence interval of [$36,021,$45,406].

These estimates represent the design-weighted average of institutional-level annual costs, accounting for the stratified sampling design and unequal sampling rates across strata. Missing cost data accounts for 44.0% of the sample overall.

Visualization